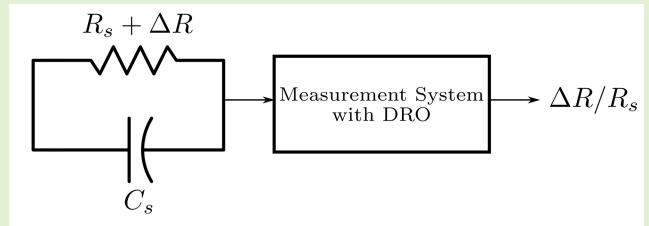


A Portable Low-Voltage Low-Power ppm-Level Resistive Sensor Measurement System

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Abstract—Objective: The objective of the paper is to present a novel systematic approach for the design and development of a portable, low-voltage low-power resistive sensor measurement system with high sensitivity. The goal of the proposed system is to be able to measure the $\Delta R/R_s$ for a wide range of baseline resistance of sensor R_s (1 k Ω to 1 M Ω i.e. 3 orders of magnitude) with high resolution and programmable sensitivity. **Method:** An auto-calibrated resistive sensor interface using a novel half-bridge based circuit is presented. The square-wave excitation and square-wave demodulation technique are utilised to reduce the power consumption of the proposed technique. Moreover, the sensitivity of the system is fully auto-calibrated for a wide range of base resistors. **Results:** With a supply voltage of 3.3 V and bandwidth of 10 Hz, the developed system is able to measure a minimum $\Delta R/R_s$ of about 20 ppm for 100 k Ω base resistance. Overall, the power consumption of the developed system is 36.3 mW (for four channels). **Significance:** The system is useful for low-voltage low-power sensor nodes as a highly sensitive readout for the resistive sensor arrays. In addition, the developed system is useful for the stain sensors in the development of flexible hybrid electronics for wearable applications.

Index Terms—Resistive sensor, readout, micro-controller, low power, auto-calibration.



I. INTRODUCTION

RESISTIVE sensors are widely used in many scientific, industrial, and agricultural applications for monitoring pressure, temperature, humidity, displacement, soil moisture, and so on. The recent development in the field of flexible resistive sensor systems leads to exciting applications such as wearable sensor systems and electronic skin to measure body temperature, and blood pressure [1], [2]. In order to integrate these flexible sensors into the subject, the system needs to be portable, low voltage and low-power.

The conventional resistance measurement system is mostly based on DC excitation [1]. However, the system based on DC excitation suffers from the thermoelectric effect due to the interconnection between the sensor, and the signal conditioning [1], [3]. This thermoelectric offset is further amplified by the instrumentation amplifier (INA) and dominantly affects the

accuracy and measurement range of the system. AC excitation is preferred to avoid the thermoelectric offset voltage [4], [5]. The sinusoidal voltage source is the first preference for the excitation signal due to its inherent single tone frequency and no additional harmonics. However, the generation of an accurate and low distortion sine wave signal is a tedious task. The oscillator-based sinusoidal signal generation consumes power since they need both active and passive components for the signal generation. In addition, the linearity in the signal generation of the oscillator can be directly related to the complexity of the oscillator [6].

The signal conditioning circuits for the resistive sensors are mostly based on the Wheatstone bridge based architecture followed by an instrumentation amplifier and a phase-sensitive-detector [7], [8]. The most commonly used phase-sensitive-detector are based on analog multiplier [7], [9]–[11]. The low-cost commercial analog multiplier consumes large power and requires a minimum 10 V for the operation (e.g. AD734, AD633, and MPY634). This limits the application of the system for low voltage low power sensing. In addition, the output of the Wheatstone bridge is non-linear with respect to the measurand [1]. Additional circuit technique is needed to linearise the output voltage, which further increases the system complexity and power consumption. The auto-nulling based signal conditioning circuits are incorporated with the bridge-based circuits for linearisation [7]. The auto-nulling based circuits require a feedback loop and

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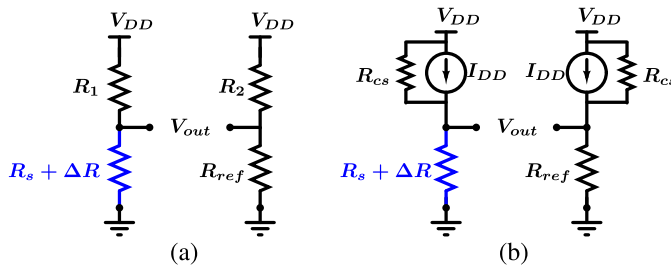


Fig. 1. (a) Wheatstone bridge (b) Current excitation bridge.

a voltage-controlled resistance element (mostly implemented using an analog multiplier and a reference resistor)—this increase the complexity and power consumption of the circuit.

Another important factor to consider while designing the signal conditioning circuit is the dynamic range of the sensor, which depends on various factors such as the measurand, fabrication process, and the sensing material. The measurement range is different for the different sensors [2], [12], [13]. The dynamic range of the conventional voltage output circuits is proportional to the supply voltage. A high supply voltage is needed for the active components of the signal conditioning circuit to measure wide-range [9], [14], [15]. The signal conditioning circuit design is especially challenging when the requirement is to measure the sensor resistance with high resolution over a wide dynamic range.

A. Comparison Between Wheatstone Bridge and Current Excitation Bridge

Wheatstone bridge and current excitation bridge are the two widely used topologies for resistive sensing applications as shown in Fig. 1. The resistance of the sensor is written as $R_s + \Delta R$, where R_s is the baseline resistance (resistance without the influence of the external stimulus) and ΔR is the incremental change in the sensor resistance of the sensor. R_{ref} is the programmable reference resistor for baseline resistor R_s compensation and R_{cs} is the output impedance of the current source.

The performance comparison of the two bridges and tabulated in Table I. The output of the Wheatstone bridge is non-linear with respect to ΔR . This limits the dynamic range of the measurement. In addition, the Wheatstone bridge performance degrades for a wide dynamic range of $\Delta R/R_s$ measurement compared to the current excitation bridge in terms of linearity, sensitivity and power supply rejection ratio (PSRR). Considering these factors, the current excitation bridge makes an excellent choice for a wide dynamic range of $\Delta R/R_s$ measurement with high resolution and programmable sensitivity.

Our group has previously reported a current excitation based half-bridge circuit for $\Delta R/R$ measurement in [4], [5]. The circuit reported in [4] is an integrated CMOS design of the high-resolution resistance measurement. The proposed circuit in this paper is an extension of the work reported in [4]. The current generator and the half-bridge circuits are modified by introducing a trans-impedance amplifier in this paper to enhance the measurement range of the sensor resistance R_s

TABLE I
COMPARISON BETWEEN WHEATSTONE BRIDGE
AND CURRENT EXCITATION BRIDGE

Parameters	Wheatstone bridge	Current excitation bridge
Output voltage (V_{out})	$V_{DD} \cdot \frac{k \frac{\Delta R}{R_s}}{(1+k) \cdot (1+k + \frac{\Delta R}{R_s})}$	$I_{DD} \cdot \frac{\Delta R}{R_s} \cdot R_s$
Sensitivity ($\frac{\partial V_{out}}{\partial (\Delta R/R_s)}$)	$V_{DD} \cdot \frac{k}{(1+k + \frac{\Delta R}{R_s})^2}$	$I_{DD} \cdot R_s$
PSRR ($\frac{\partial V_{DD}}{\partial V_{out}}$)	$\frac{(1+k) \cdot (1+k + \frac{\Delta R}{R_s})}{k \frac{\Delta R}{R_s}}$	$\frac{k}{\frac{\Delta R}{R_s}}$
k	$\frac{R_1}{R_s} = \frac{R_2}{R_{ref}} = 1$	$\frac{R_{cs}}{R_s} > 1000$
Max Sensitivity at $\Delta R/R_s = 0$	$V_{DD}/4$	$I_{DD} \cdot R_s = V_{DD}/2$ #
PSRR at $\Delta R/R_s = 100\%$	15 dB	60 dB
Current Consumption	V_{DD}/R_s	$2 \cdot I_{DD} = V_{DD}/R_s$ #
Prog. Sensitivity for different base resistor	3 prog. resistor & prog. V_{DD}	1 prog resistor & prog. I_{DD}

Considering the same power supply and AC excitation $I_{DD} \cdot R = V_{DD}/2$

with lower supply voltage V_{DD} . The salient features of the proposed system are as follows:

1) **Wide Measurement Range:** A novel active half-bridge based topology is used for converting the current through the sensor resistance into an equivalent voltage signal. A reference element is used in the half-bridge circuit to obtain the differential operation in order to compensate for the baseline resistance of the sensor.

2) **Low Power:** To reduce the power consumption, the proposed system utilises square wave current excitation, which can be easily generated using a low-power micro-controller. In addition, a synchronous demodulator is proposed with a single-ended INA output.

3) **Accuracy:** The effect of multiple harmonics at the output due to the square wave excitation signal, the effect of chopping spikes and the effect of power supply interference are compensated in the digital domain using a moving average filter. The null points of the moving average filter are selected as the greatest common divisor (GCD) of such frequencies. This significantly improves the accuracy of the measurement.

4) **Auto-Calibration:** The system is auto-calibrated to provide high sensitivity. The flexibility is provided by auto-programming the gain of the INA and the amplitude of the current source to obtain the maximum sensitivity in the measurement of the sensor resistance. Additionally, a differential and ratio-metric operation is performed on the acquired voltage signal, which reduces the effect of the mismatch and further improves the accuracy of the measurement.

II. PROPOSED MEASUREMENT SYSTEM

The schematic diagram of the proposed system is shown in Fig. 2. The operating principle of the proposed system is based on a current-to-voltage conversion followed by a differential operation, synchronous demodulation, and filtering. A current flows through a sensor and a reference branch, and the differential input voltage is converted to single-ended INA output, which is synchronously demodulated and filtered to obtain the $\Delta R/R_s$ of the resistive sensor. The sensor resistance

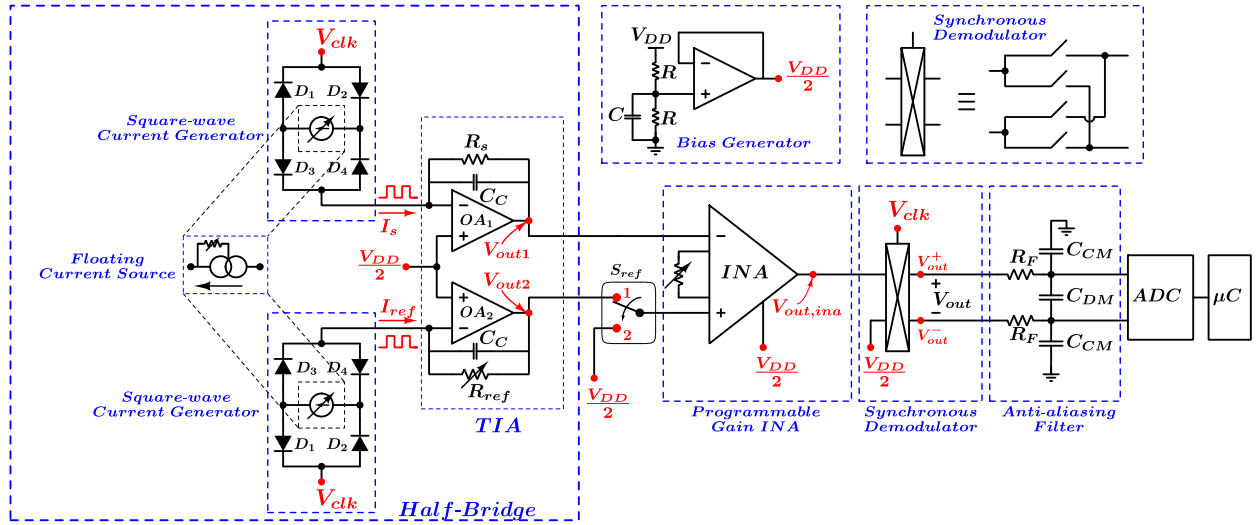


Fig. 2. The circuit diagram of the complete $\Delta R/R_s$ measurement system.

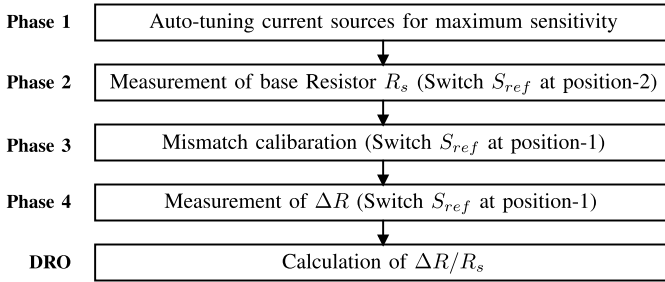


Fig. 3. Flowchart describing the operation of the proposed resistance measurement system.

is represented by R_s , and the reference resistor is represented by R_{ref} (a programmable digital potentiometer).

The reference resistor R_{ref} is auto-configured using a micro-controller to compensate the baseline resistor of the sensor at the beginning of the measurement cycle when the sensor is not under the influence of the stimulus. The gain of the INA and the Programmable current sources are auto-programmed to obtain the maximum sensitivity for the measurement of ΔR .

The four-phase auto-calibration operation [4] of the system is illustrated in the flowchart shown in Fig. 3. The details about the four-phase $\Delta R/R$ measurement is available in [4].

The system consists of a proposed square-wave current generator, a proposed active half-bridge based circuit for converting the current into voltage, an instrumentation amplifier with gain control, an analog switch based synchronous demodulator, an anti-aliasing filter, an analog-to-digital converter, and a low power microcontroller, as shown in Fig. 2.

A. Proposed Excitation Current-Generator

The schematic diagram of the conventional current generator and the proposed current generator is shown in Fig. 4(a) and (b), respectively. The programmable floating current source (PFCS) is implemented using an adjustable

current source IC, a diode bridge network, and a digital potentiometer.

The bi-directional current is generated as per the potential V_{AB} across the nodes A & B. When V_{clk} is high, the diodes D_2 and D_3 are forward bias, and the current flow through the nodes A, C, D, and B and dumped into the sensor resistor R_s . When V_{clk} is low, the diodes D_1 and D_4 are forward bias, and the current flows in the reverse direction from node B to C, D, and A.

The conventional current generator [5], [16], [17] as shown in Fig. 4 (a). In the conventional approach, the sensor resistor is connected across nodes B & E. The condition for the current generator to operate properly is:

$$|V_{AB}| > (2 \cdot V_F + V_{comp}) \quad (1)$$

where V_F is the cut-in voltage of the diodes, V_{comp} is the minimum operating voltage required across the current source IC and I_{FCS} is the current generated due to the programmable floating current source. The potential at node B depends on the value of I_{FCS} and sensor resistor R_s . The higher values of I_{FCS} and R_s increase the voltage at node B. This limits the headroom for the voltage $|V_{AB}|$, and the condition of current generator Eq. (1) needs to be considered carefully. This eventually limits the range of the sensor resistance measurement. The maximum sensor resistor $R_{s,max}$ that can be used with the conventional current generator as shown in Fig. 4(a) is given as follows.

$$R_{s,max} = \left(\frac{\frac{V_{DD}}{2} - 2V_F - V_{comp}}{I_{FCS}} \right) \quad (2)$$

The proposed current generator, as shown in Fig. 4(b), solves the problem as mentioned above by introducing an operational amplifier (opamp), which works as a trans-impedance amplifier. The negative feedback around the opamp forces the voltage at node B to reference level $V_{DD}/2$. Therefore, the potential at node B is independent of the sensor resistance. This enhances the range of sensor resistance measurement compared with the conventional approach. The maximum

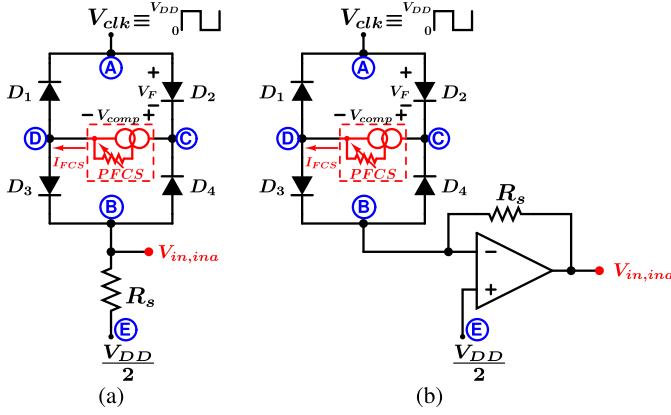


Fig. 4. The circuit diagram of the programmable current generator (a) conventional (b) proposed.

TABLE II
COMPARISON OF CONVENTIONAL AND PROPOSED CURRENT GENERATOR IN-TERMS OF MAXIMUM BASE RESISTOR ($R_{s,max}$) AT 3.3 V SUPPLY

Component	Parameter	Value	$R_{s,max}$		Improvement
			Conventional	Proposed	
LM334	$I_{FCS,min}$	1 μ A	250 k Ω	1.65 M Ω	$\times 6.6$
	V_{comp}	1 V			
UC3611	V_F	0.2 V			
LP5907	V_{DD}	3.3 V			

sensor resistor $R_{s,max}$ that can be used with the proposed current generator is given as follows.

$$R_{s,max} = \left(\frac{V_{DD}}{2 I_{FCS}} \right) \quad (3)$$

Table II illustrate the improvement in R_s measurement range using the proposed current generator for a supply voltage of 3.3 V. The proposed generator helps to achieve a 6.6 times more comprehensive measurement range than the conventional approach. The other interesting feature of the proposed current generator is the ability to tackle the effect of induced supply noise. Since one terminal of the sensor is connected to the virtual ground and the other is at the low impedance node, the variation in the voltage due to induced noise will be negligible.

B. Active Half-Bridge

As mentioned previously, the current generated using the current generators I_s & I_{ref} flows through R_s & R_{ref} and generate the voltage V_{out1} and V_{out2} , respectively, as shown in Fig. 2. The voltage V_{out1} and V_{out2} is applied to an instrumentation amplifier (INA) to generate a voltage $V_{out,ina}$, which is proportional to difference between the resistance R_s and R_{ref} . The expression for the voltage $V_{out,ina}$ can be written as follows.

$$V_{out,ina} = \frac{V_{DD}}{2} + A \times [(I_s(t) \times R_s) - (I_{ref}(t) \times R_{ref})] \quad (4)$$

where A is the gain of the INA. Considering the current $I_s(t) = I_{ref}(t) = I(t)$, the expression for the voltage $V_{out,ina}$

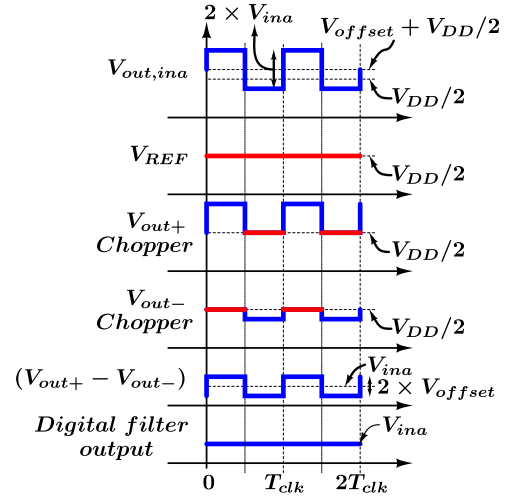


Fig. 5. Synchronous Demodulation with single-ended INA output in time domain.

can be modified as follows.

$$V_{out,ina} = \frac{V_{DD}}{2} + A \times I(t) \times [R_s - R_{ref}] \quad (5)$$

C. $\Delta R/R$ Measurement

The primary reason to perform the ratio-metric operation was to compensate the additional DC term D in the output due to the square wave current injection and square wave for the synchronous demodulation. The term D cancels out due to the ratio-metric operation. Along with the DC term cancellation, the ratio-metric operation also helps in compensating the effect of variation of outside conditions such as power supply variance and many others. The ratio-metric operation was performed between the differential output voltage $V_{out}|_{\text{phase 4}} - V_{out}|_{\text{phase 3}}$ and the output voltage $V_{out}|_{\text{phase 2}}$. The expression for the differential and ratio-metric operation can be given as follows.

$$DRO = \frac{V_{out}|_{\text{phase 4}} - V_{out}|_{\text{phase 3}}}{V_{out}|_{\text{phase 2}}} \quad (6)$$

$$= \frac{\Delta R}{R_s} \quad (7)$$

Eq. (7) shows that the four-phase differential and ratio-metric measurement provides output proportional to $\Delta R/R_s$. This compensates for the effect of current mismatch δI , the initial mismatch δR between the R_s and R_{ref} , and the effect of harmonics on the measurement of sensor resistance.

III. ACCURACY ANALYSIS

Apart from the mismatches in the current sources and resistors, the accuracy of the system is also affected by several other factors. These factors include the noise of the current source, noise of the opamps and INA, offset and offset drift of INA, chopping artifacts and supply noise interference. The effect of these noise sources on the output and the methods utilised to compensate for the effect of these noise sources are described in this section.

A. Parasitic Capacitance

Most of the resistive sensors have a parasitic capacitance in parallel with the sensor resistance. This parasitic capacitance may affect the performance of the measurement system and therefore needs to be taken into consideration.

The output voltage V_{out} after digital filtering considering the parasitics capacitors $C_{s,par}$ and $C_{ref,par}$ across R_s and R_{ref} , respectively, can be expressed as follows.

$$V_{out} = \left(\frac{4}{\pi}\right)^2 \frac{A.I}{2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \left(\frac{R_s}{1 + (2n-1)^2 \omega_{clk}^2 \cdot R_s^2 \cdot C_{s,par}^2} - \frac{R_{ref}}{1 + (2n-1)^2 \omega_{clk}^2 \cdot R_{ref}^2 \cdot C_{ref,par}^2} \right) \quad (8)$$

The expression (8) shows that the output voltage is affected by the parasitic capacitances. This affects the accuracy of the measurement of $\Delta R/R_s$. Therefore, some consideration must be made to reduce the effect of the parasitic capacitances.

The expression (8) is modelled to obtain the error in the measurement of the sensor resistance for different values of the excitation frequency ω_{clk} .

The condition for the compensation of the error due to the parasitic capacitances on the sensor resistance measurement can be written as follows.

$$\omega_{clk} \ll \frac{1}{R_s C_{s,par}} \quad \text{and} \quad \omega_{clk} \ll \frac{1}{R_{ref} C_{ref,par}} \quad (9)$$

Considering the above conditions, the expression (8) can be simplified as follows.

$$V_{out} \approx \left(\frac{4}{\pi}\right)^2 \frac{A.I}{2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} (R_s - R_{ref}) \quad (10)$$

Therefore, the effect of the parasitic capacitance can be reduced significantly by using lower excitation signal frequency [4].

B. Effect of Chopping Artifacts, High Order Harmonics, and Supply Noise (50/60 Hz)

The primary motivation while designing the system is to be able to develop a low-voltage, low-power $\Delta R/R_s$ measurement system. The system utilises a square-wave signal for excitation as well as demodulation. Therefore, the final output of the circuit is affected by factors like chopping spikes and artifacts. The chopping of square wave signals produces chopping artifacts due to the finite bandwidth of the instrumentation amplifier. These artifacts degrade the resolution and accuracy of the system. In addition, the effect of the higher-order harmonics such as $2f_{clk}$, $4f_{clk}$, also needs to be removed for accurately measuring the $\Delta R/R_s$.

The chopping artifacts and chopping spikes occur at a frequency that is multiple of the excitation frequency. The effect of the chopping spikes, harmonics and supply noise can be eliminated from the output voltage by using a moving average filter. The null points of the moving average filter (f_s/N)

can be selected at a frequency that is the greatest common divisor (GCD) of the frequency of the excitation frequency, chopping spikes frequency and supply noise frequency.

$$f_{null} = \frac{f_s}{N} = \text{GCD}(f_{chop}, n f_{chop}, f_{supply_noise}) \quad (11)$$

C. Effect of Input Offset Voltage and Bias Current

The non-idealities of the circuit components affects the performance of the proposed system. In this section, we analyzed the effect of various circuit parameters on the output of the circuit. The following non-idealities are considered:

- The input offset voltage of operational amplifier OA_1 and OA_2 as V_{off1} and V_{off2} , respectively.
- The input offset voltage of the instrumentation amplifier INA as V_{off3} .
- The bias current of operational amplifier OA_1 and OA_2 as I_{b1} and I_{b2} , respectively.
- The thermoelectric voltage generated due to the interface between sensor and electronics as V_{off_sen} .

Considering these, the expression of the output voltage V_{out} can be written as follows.

$$V_{out}(t) = \underbrace{\left(\frac{4}{\pi}\right)^2 \times A \cdot [I \cdot (R_s - R_{ref}) - \delta I \cdot R_{ref}] \times [H(t)]^2}_{Desired} + \underbrace{\left(\frac{4}{\pi}\right) \cdot A \cdot [V_{offset}] \times H(t)}_{Error} \quad (12)$$

where $V_{offset} = V_{off1} + V_{off2} + V_{off3} + I_{b1}^- R_s + I_{b2}^- R_{ref} + V_{off_sen}$

The effect of input offset voltage is represented by the error term in Eq. 12. This error term is analysed further to understand the effect of offset on the output voltage V_{out} . The error term in the frequency domain after the moving average filter can be written as follows.

$$V_{e_filter}(f) = \left(\frac{4}{\pi}\right) [V_{offset}] \cdot f_s \cdot A \cdot H(e^{j2\pi f/f_s}) \cdot \sum_{k=-\infty}^{\infty} H(f - k \cdot f_s) \quad (13)$$

From (13), the error due to the offset voltage can be reduced to zero by carefully choosing the null points of the moving average filter. The above expression can be reduced to zero if it satisfies the following condition.

$$H(e^{j2\pi f/f_s}) \cdot \sum_{k=-\infty}^{\infty} H(f - k \cdot f_s) = 0 \quad (14)$$

The expression (14) can be further expanded as follows.

$$\begin{aligned} & \frac{1}{N} \frac{\sin[(2\pi f/f_s) \cdot N/2]}{\sin[(2\pi f/f_s)/2]} \sum_{k=-\infty}^{\infty} \sum_{n=1}^{\infty} \\ & \times \frac{\delta[f - kf_s - (2n-1)f_{clk}] - \delta[f - kf_s + (2n-1)f_{clk}]}{2j(2n-1)} \\ & = 0 \end{aligned} \quad (15)$$

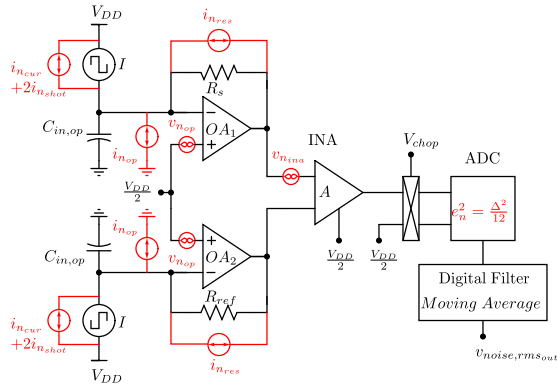


Fig. 6. Noise sources in the proposed system.

Using the above expression, the conditions $f_s = N \cdot f_{clk}$ and $N = (\text{even})$ need to be satisfied to eliminate the effect of offset. (16), as shown at the bottom of the page.

The expression verifies that with the careful design of the moving average filter, the errors in the $\Delta R/R_s$ measurement due to the input offset voltage & input bias current of opamps & INA, and chopping spikes & artifacts can be removed.

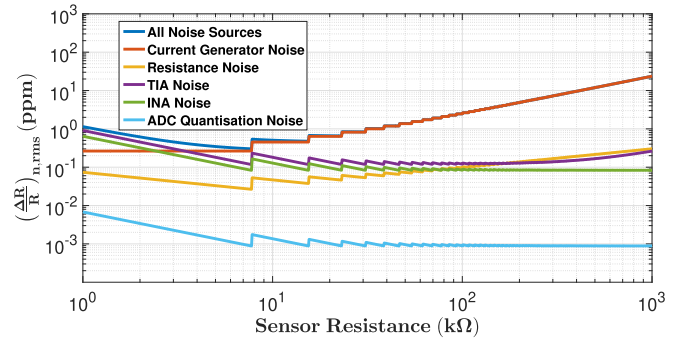
D. Effect of Noise Due to Components

The measurement resolution of the $\Delta R/R_s$ is dependent on the noise of the system. The noise sources affecting the resolution are current source noise, diode shot-noise, opamp voltage noise and current noise, TIA noise due to parasitic capacitor C_p , INA noise voltage and the quantisation noise of the ADC. Considering the input-referred noise voltage due to operational amplifiers (OA_1 , OA_2) and instrumentation amplifier INA as $V_{n,op}$ and $V_{n,ina}$, respectively. The noise due to the current source as $i_{n,cur}$ and the operational amplifier current noise as $i_{n,op}$. The shot noise due to the diodes is $2qI_D$. The expression for the output voltage $V_{noise,rms,out}$ can be expressed as.

$$\Delta R_{noise,rms} = \frac{v_{noise,rms,in}}{I_{max}} = \frac{v_{noise,rms,out}/A}{\min \left\{ \frac{V_{ICMR,ina}/2}{R_s}, I_{FCS} \right\}} \quad (17)$$

$$\left(\frac{\Delta R}{R_s} \right)_{noise,rms} = \frac{v_{noise,rms,out}/A}{\min \left\{ V_{ICMR,ina}/2, I_{FCS} \cdot R_s \right\}} \quad (18)$$

The effect of the different component noise sources on $\Delta R/R_s$ measurement is plotted in Fig. 7. The $\Delta R/R_s$ of the circuit is mostly affected by the noise of the current source. Overall, the worst-case $\Delta R/R_s$ due to the cumulative effect of noise sources is less than 20 ppm for a baseline resistance

Fig. 7. $\left(\frac{\Delta R}{R_s} \right)_{n,rms}$ (ppm) Vs Sensor Resistance Noise contribution of individual components.

of 100 kΩ. The proposed system's analytical expression for the noise is similar to the circuit reported in [4]. The current source in [4] is affected by the flicker noise of CMOS transistors. On the other hand, the current source in this paper is implemented using LM334, a BJT-based current source with zero flicker noise. Therefore, the performance of the system reported in this paper is comparable with [4] despite being implemented using the discrete components.

IV. EXPERIMENTAL SETUP AND RESULTS

A prototype board of the proposed circuit was designed and tested in the laboratory. The prototype was designed for four-channel $\Delta R/R_s$ measurement. The prototype consists of sensor resistors, current generators for 4-channel half-bridge, instrumentation amplifiers, a switch for synchronous demodulation, a 24-bit ADC, a micro-controller for logic and control, a Bluetooth module for wireless communication and a power management unit.

Each channel consists of a bi-directional current generator. The LM334 adjustable current source along with a 100 kΩ digital potentiometer (AD5242BRUZ100 from Analog Devices) is used for the programmable current source. The DC current generated using LM334 is converted into square-wave using diode-bridge. The diodes in the current generator need to be matched to obtain a symmetric square-wave current (equal sourcing and sinking). Symmetric square-wave current is implemented using matched quad switching diodes (UC3611 by Texas Instruments). The remaining components model and values are presented in Table-III.

The micro-controller is used for control and logic implementation as well as for current source calibration. The current source was calibrated once the desired voltage across the opamp output was obtained. The output of the synchronous demodulator is acquired using a 24-bit ADC (ADS131A04,

$$v_{noise,rms,out}^2 = \int_0^{\frac{f_s}{2}} A^2 \cdot \left(\underbrace{\frac{4 \cdot (2qI_D) \cdot R^2 + 2 \cdot (\overline{i_{n,cur}^2} \cdot R^2)}{2 \cdot \overline{v_{n,op}^2} \left(\frac{2\pi f - 3dB C_{in,op}}{3} \right)^2}}_{\text{Current Generator Noise}} + \underbrace{\frac{2 \cdot (4kTR)}{2 \cdot \overline{v_{n,op}^2} \left(\frac{2\pi f - 3dB C_{in,op}}{3} \right)^2}}_{\text{Resistor Noise}} + \underbrace{\frac{\overline{v_{n,ina}^2}}{2 \cdot \overline{v_{n,op}^2} \left(\frac{2\pi f - 3dB C_{in,op}}{3} \right)^2}}_{\text{INA Noise}} + \underbrace{\frac{\Delta^2 \cdot 1}{12 \cdot A^2 \cdot f_s/2}}_{\text{ADC Noise}} \right) \cdot |H_{mov_avg}(f)|^2 \cdot df \quad (16)$$

TABLE III
COMPONENTS

Component	Model
Schottky diode (D)	UC3611
Floating Current Source (I)	LM334
Opamp (TIA, $V_{DD}/2$ generator)	OPA2333
Instrumentation Amplifier (INA)	INA333
100k Ω Digital Potentiometer (R_{INA}, R_{CG})	AD5242BRUZ100
1M Ω Digital Potentiometer (R_{ref})	AD5242BRUZ1M
SPDT analog switch (S_{ref})	TS3A24159
Synchronous Demodulator (Switches)	TS3A44159
Micro-controller	MSP430F5529
24 bit ADC	ADS131A04
Digital Buffer	SN74LVC2G34
3.3 V LDO	LP5907
Li-ion Battery Charger with Power Management	BQ24078

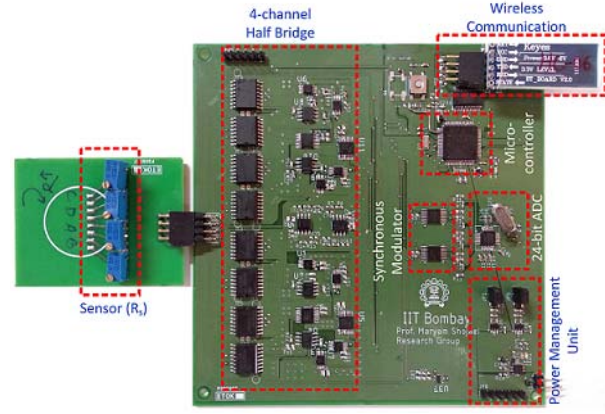


Fig. 9. System prototype.

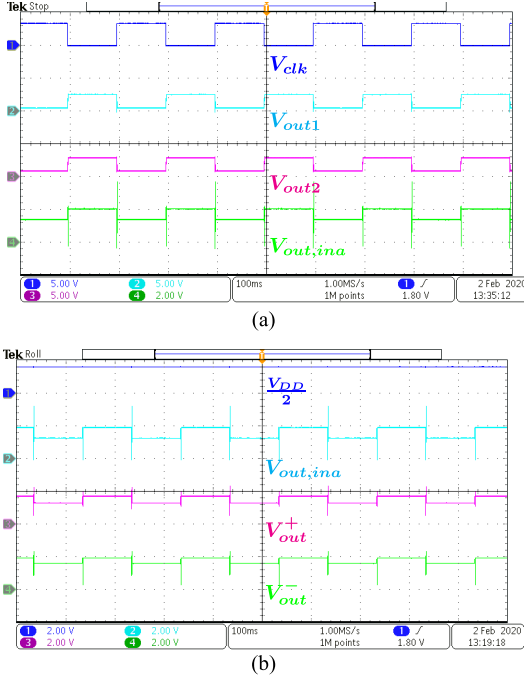


Fig. 8. (a) Signals at modulation nodes (b) Signals at demodulation nodes in Fig. 2.

from Texas Instruments). The ADC is oversampled to reduce the noise of the acquired signal. The synchronous demodulator is implemented using a switch (TS3A44159, from Texas Instruments). The system is provided with wireless communication capability with an onboard Bluetooth module. A power management module is designed for low noise and battery-powered operation.

A. $\Delta R/R_s$ Measurement

In this experiment, the performance of the developed system is tested for minimum $\Delta R/R_s$ measurement. The experiment is conducted for the worst-case measurement scenario when the baseline resistance is highest. The proposed system is designed for the maximum R_s of 1 M Ω . One SMD resistor is used to set the value of the baseline resistance R_s and one trimmer potentiometer is connected in series to emulate the sensor for each channel for the ΔR variation, as shown in Fig. 9. In this experiment, the 500 k Ω (25 turns) trimmer

TABLE IV

EXPERIMENTAL RESULTS OF TRANSIENT RESPONSE OF THE SYSTEM WITH DIFFERENT VALUES OF BASELINE RESISTOR R_s

Base resistor R_s	Emulated $\Delta R^{\#}$	1 LSB $^{\#}$ $\Delta R/R_s$	Range $^{\#}$ ppm	Max Relative Error in step
1 k Ω	1 k Ω	3900 ppm	0 – 19.5k	0.87%
10 k Ω	1 k Ω	390 ppm	0 – 1950	–0.64%
100 k Ω	10 k Ω	390 ppm	0 – 1950	0.8%
1 M Ω	100 k Ω	390 ppm	0 – 2730	0.71%

$^{\#}$ 8-bit, 20% tolerance digipot (Analog Devices AD5242)

potentiometer is used for emulating ΔR . The utilised trimmer potentiometer has a tolerance of $\pm 10\%$. The tolerance is taken into consideration by first calibrating the Digi-pot with a precision LCR meter (E4980 from Agilent Technologies). The experimental results show that the system is able to detect a minimum $\Delta R/R_s$ of 20 ppm for a baseline resistance of 100 k Ω . The error in the measurement of $\Delta R/R_s$ is within $\pm 1\%$.

B. Measurement of $\Delta R/R_s$ for Wide-Range of R_s Values

The accurate measurement of the minimum $\Delta R/R_s$ value highly depends on the value of the baseline resistor R_s . The noise is proportional to the value of R_s , which affects the lower limit of the measurement of $\Delta R/R_s$ accurately. In this experiment, the value of $\Delta R/R_s$ for different values of the R_s is measured.

First, the R_s is fixed at one value, and the ΔR is varied in steps using the digital-potentiometer. The value of $\Delta R/R_s$ is measured, and the percentage relative error is calculated. Next, the same process is repeated for different values of R_s . The results of the experiment are tabulated in Table IV. The base resistance of the sensor is varied from 1 k Ω to 1 M Ω . The value of ΔR is varied in step using the Digi-pot. The maximum percentage relative error for the range of $\Delta R/R_s$ is also reported in Table V. The result shows that the developed resistive measurement system is able to detect a wide range of sensor resistance with an error of less than $\pm 1\%$.

C. Other Performance Parameters

The other performance parameters, such as the power consumption, maximum relative error and the standard deviation

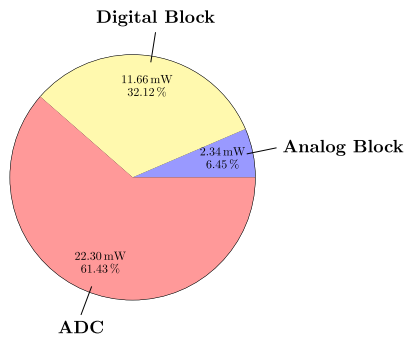


Fig. 10. Power distribution of digital block (μC + clock buffer + synchronous demodulator), Analog block (TIA + Current generator + INA) and ADC.

TABLE V

EXPERIMENTAL RESULTS WITH DIFFERENT BASE RESISTOR R_s

Base Resistor R_s	$\Delta R^\#$	Measured $\Delta R/R_s$ (ppm)	Actual $\Delta R/R_s$ (ppm)	Relative Error	Std. dev
1 k Ω	3.9 Ω	3481	3453	0.81%	4
10 k Ω	3.9 Ω	347.4	345.3	0.6%	2.3
100 k Ω	39 Ω	356	352.6	0.96%	3.1
1 M Ω	390 Ω	380.8	383.5	-0.7%	18

8-bit, 20% tolerance digipot (Analog Devices AD5242)

of the developed $\Delta R/R_s$ measurement system, is experimentally measured. The power consumption of the developed $\Delta R/R_s$ system is around 36.3 mW. The power contributed by different modules of the developed system are shown in Fig. 10. The standard deviation of the developed $\Delta R/R_s$ system for different values of the base resistor and ΔR are tabulated in Table V. The results show that the proposed system is able to detect a wide range of $\Delta R/R_s$ with excellent resolution and standard deviation.

D. Measurement of $\Delta R/R_s$ of Strain Sensors With the Proposed System

The performance of the proposed $\Delta R/R_s$ measurement system is further tested with strain sensors. The details of the sensor fabrication are reported in [4]. The fabricated system is tested with three strain sensors. The $\Delta R/R_s$ value of the strain sensors at different values of deflection is measured and shown in Fig. 11. The result shows that, despite implementing using the commercial-off-the-shelf-components, the performance of the system is similar to our previous reported system implemented using CMOS technology.

V. PERFORMANCE COMPARISON AND DISCUSSIONS

The performance of the proposed $\Delta R/R_s$ measurement system is compared with the other work reported in the literature and presented in Table VI. The features and performance parameters of the developed system are compared with the other $\Delta R/R_s$ systems. The summary of Table VI is as follows.

- 1) Compared to other systems, the proposed system utilises programmable current source and INA gain,

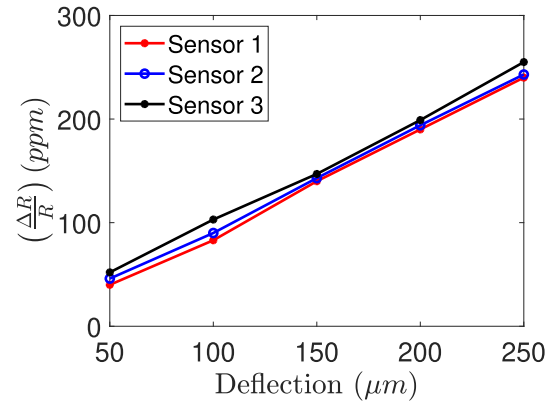


Fig. 11. Experimental results of strain sensors with the $\Delta R/R_s$ measurement system.

which helps in auto-calibrating the system for baseline resistance compensation and high sensitivity $\Delta R/R_s$ measurement. The wide range of the excitation current signal helps in a wide range of resistance measurements.

- 2) The proposed system consumed little power compared to the other reported work. The proposed system runs on 3.3 V, which is significantly less than the power supply used by all the reported systems in Table VI. The power consumption of the proposed system (for four channels) is significantly lower than the other systems. Thanks to the low supply voltage and small power consumption, the proposed system is suitable for the portable $\Delta R/R_s$ applications. It's worth mentioning that the low-voltage low-power design of the proposed system is possible due to the novel square-wave excitation and demodulation approach.
- 3) Even with the low-supply voltage, the system is able to detect a minimum $\Delta R/R_s$ of about 20 ppm with base resistor R_s of 100 k Ω . This is a very useful feature of the proposed system. Overall, the proposed system is a one board solution and does not require calibration for the $\Delta R/R_s$ measurement, unlike other reported works.
- 4) The bandwidth of the proposed system is significantly higher than most of the other reported works in Table VI. Reducing the bandwidth reduces the noise, and hence, a lower $\Delta R/R_s$ measurement can be achieved, whereas the response time increases. Measurement of $\Delta R/R_s$ with high bandwidth is challenging. Therefore, for the fair comparison between the systems, a figure of Merit (FOM) is defined, which includes power dissipation, bandwidth (response time) and minimum measurable $\Delta R/R_s$.

$$FOM(W/\sqrt{Hz}/ppm) = \frac{\Delta R}{R} \cdot \frac{V_{DD} \cdot I}{\sqrt{\min\{f_{clk}/2, BW\}}} \quad (19)$$

According to the above FOM, the proposed system provides excellent performance compared to the other system reported.

TABLE VI
COMPARISON OF THE PROPOSED WORK WITH THE OTHER PUBLISHED WORK

Channel(s)	This work	[17]	[18]	[19]	[20]	[21]	[4]
Excitation Signal	Current	Current	Voltage	Voltage	Current	Voltage	Current
Base Resistor R_s	1 k Ω – 1 M Ω	100 Ω – 4 M Ω	10 k Ω – 1 M Ω	1 k Ω – 1 M Ω	8 k Ω – 230 k Ω	56 k Ω – 6.5 M Ω	1 k Ω – 500 k Ω
Excitation Current Range	Programmable 1 μ A – 170 μ A	Fixed 1 μ A	-	-	Fixed 50 μ	-	Programmable 0.1 μ A – 44.4 μ A
INA Gain	Programmable 1 – 257	Fixed	Fixed	-	Fixed	-	Fixed
Measurement	$\Delta R/R$	ΔR	ΔR	ΔR	R	R	$\Delta R/R$
Supply	3.3 V	± 5 V	± 12 V	$> \pm 5^{\#}$ V	± 15 V	± 7 V	1.8/3.3 V
Power	36.3 ^ψ mW	420 mW	720 mW	NA	NA	97.3 mW	NA
Bandwidth	10 Hz	4 Hz	0.48 Hz	0.35* Hz	NA	1.4* Hz	10 Hz
Min $\Delta R/R$	20 ppm @ $R_s = 100$ k Ω	150 [†] ppm @ $R_s = 100$ k Ω	6 ppm @ $R_s = 110$ k Ω	100 ppm @ $R_s = 100$ k Ω	35 ppm @ $R_s = 100$ k Ω	323 ppm @ $R_s = 100$ k Ω	48 [†] ppm @ $R_s = 100$ k Ω
Sensor Excitation	On-board	On-board	External	On-board	On-board	External	On-board
Auto Calibration	Yes	Yes	No	Yes	No	No	Yes
Programmable Sensitivity	Yes	No	No	Yes	No	No	Yes
Thermoelectric EMF Mitigation	Yes	Yes	Yes	Yes	No	Yes	Yes
Parasitics Considerations	Yes	No	No	Yes	-	Yes	Yes
FOM [§] (W/ $\sqrt{Hz/ppm}$)	0.06	7.9	6.24	NA	NA	26.5	NA

Supply voltage not reported

* Calculated from reported response time

† Calculated from reported sensitivity

ψ Wireless module power not included

§ FOM per channel

VI. CONCLUSION

A low-voltage low-power high resolution $\Delta R/R$ measurement system with a novel current source is reported in this paper. The system is based on a half-bridge-based circuit with square-wave current excitation. The circuit is implemented using commercial-off-the-shelf components and powered with 3.3 V. The experimental results show that the proposed circuit can measure a wide range of sensor resistance (1 k Ω - 1 M Ω) with a minimum $\Delta R/R$ resolution of 20 ppm. The bandwidth and power consumption of the system was 10 Hz and 36.3 mW for four-channel measurement, respectively. Overall, thanks to the square-wave current excitation and novel current source, the proposed method, implemented using discrete components, is able to provide performance similar to over previously reported CMOS implemented $\Delta R/R$ system [4].

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